

Qiskit v2.X Developer Certification

Study group kick-off



Astri Cornish
IBM Quantum

Radha Pyari Sandhir
IBM Quantum



Agenda

Qiskit v2.X Certification

Why create a study group?

Best practices for group study

Effective communication strategies

Expectations and managing conflicts

How to form a study group

Active participation

Resources

Q&A

IBM Certified Quantum Computation using Qiskit v2.X Developer - Associate

The [“Qiskit v2.X Certification,”](#) for short, is a recognised professional certification that proves an individual possesses a fundamental knowledge of quantum computing concepts and can demonstrate this using the Qiskit SDK as well as Qiskit Runtime.

Why obtain this certification?

- Anyone with the new Qiskit developer certification is considered **an IBM-certified quantum computational scientist**.
- **Certifications are widely recognized** and highly respected across the global IT industry.
- **Be one of ~150 people globally** who have this certification
- Makes a real impact for anyone looking to **advance their career** in quantum computation.
- Gaining up-to-date skills in Qiskit will **improve your impact on the Qiskit community** - this certification is a hurdle for **progressing to Tier 1** in the Qiskit advocate program.

Qiskit v0.2X (old!) certification:
1300+ certified Qiskit developers
across 71 countries



Fundamentals of Quantum Computing Using Qiskit v2.X Developer exam

To obtain the certification, you must pass a proctored exam
“Fundamentals of Quantum Computing Using Qiskit v2.X Developer” with
the following topics:

- Perform quantum operations
- Visualize quantum circuits, measurements, and states
- Create quantum circuits
- Run quantum circuits
- Use the sampler primitive
- Use the estimator primitive
- Retrieve and analyze the results of quantum circuits
- Operate with OpenQASM

More information about the certification and exam:

- [Blog](#)
- [Advocate seminar](#)

Number of questions: 68
Number of questions to
pass: 47

Time allowed*: 90 minutes

*Test accommodations are
available

Why create a study group?

The Qiskit v2.X Certification exam is rigorous.

Studying alone => hard to stay consistent

Study groups will give you:

- Peer accountability and motivation
- Shared knowledge and problem solving
- Deeper understanding (not just enough of an understanding to pass the exam)
- Stronger Advocate community bonds
- Advocate points and a voucher to take the exam for free!



Best practices for group study

- **Set clear goals and objectives**
 - Overall goal: "Obtain the Qiskit v2.X Certification by ____"
 - Other goals in the group? Why earn the certification?
- **Create a study structure and schedule with *regular* touch-points**
 - Understand how each member in the group will most effectively learn and engage in the study group
 - Create week-to-week learning goals
- **Encourage active participation amongst the whole group and maintain open communication**
- **Utilize existing and new learning resources**
 - Resources from IBM Quantum, QAP, Qiskit community
 - Keep a shared doc with notes, track open questions...

Study group structure

Group size: 3-6 people

Duration: 12 weeks

Recommended Execution Structure

Cadence: weekly 60-90 min sessions

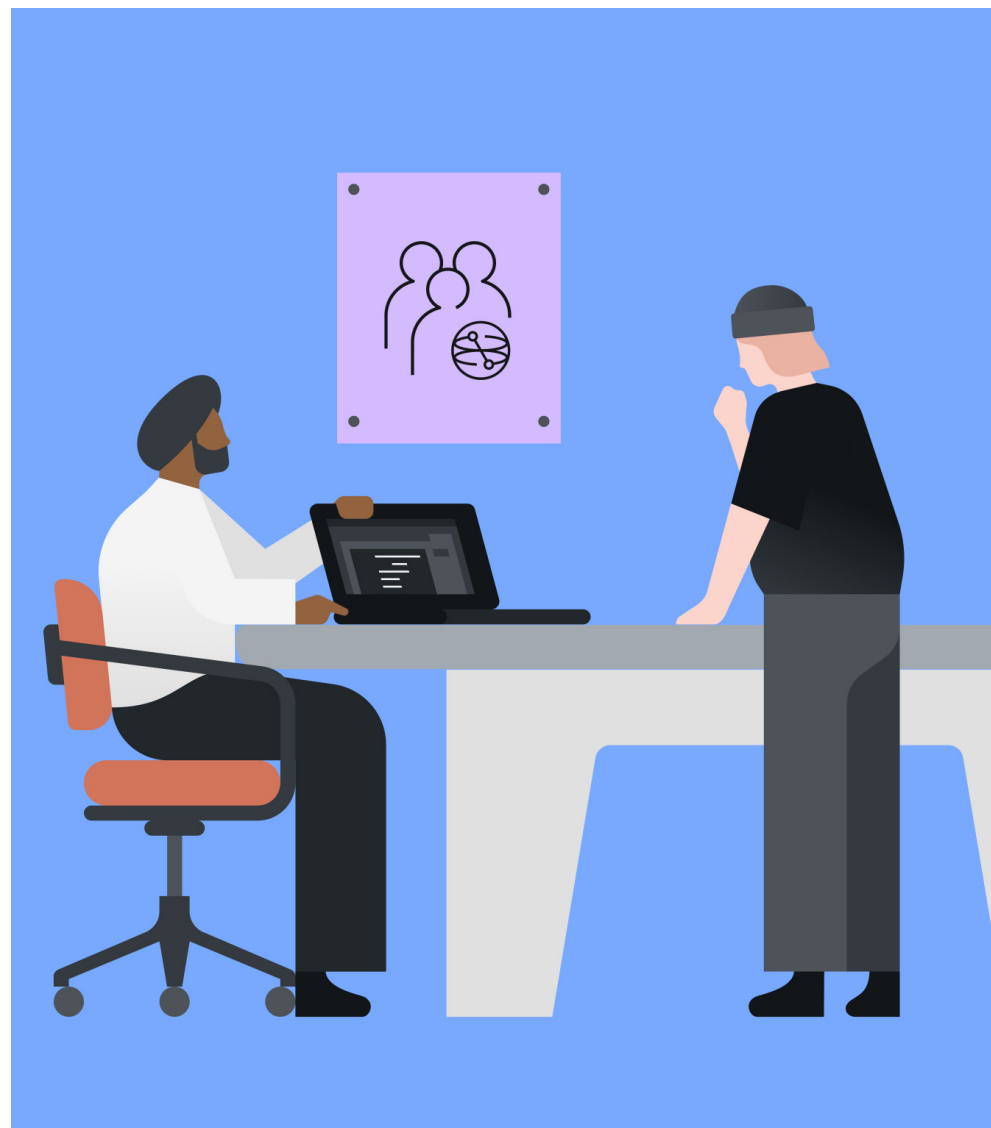
Suggested weekly meeting agenda:

1. Concept review (15-20 mins)
2. Hands-on coding (30-40 mins)
3. Practice questions (15-30 mins)
4. Set tasks for group to complete before next session

Rotate facilitators each week where someone in the group leads the meeting.

Effective communication strategies

- Find a regular meeting time that suits *everyone* in the group
 - Communicate *early* if you cannot join a meeting as a once-off occasion
 - If you can no longer keep up with the study group, discuss with your group that you're finding it hard to keep up, and communicate clearly if you need to leave the study group for some reason.
- Be constructive and curious
 - Critique ideas rather than people
 - e.g. "Can you explain your reasoning?" not "that's wrong"
- Be inclusive:
 - Encourage all group members to ask questions and contribute to discussions
 - Make space for quieter members and avoid dominating discussions
 - Clearly define actions, roles, and next steps for each person
- Utilize collaboration tools that everyone has access to
 - e.g. GitHub for shared coding, Google Docs for keeping group notes, Discord for async group chats.



Expectations and managing conflicts

Common sources of conflict:

- Different skill levels
- Ambiguity in role definitions
- Uneven preparation
- Dominating personalities

– Address conflicts early:

- Early communication of friction will prevent them from escalating.
- If relevant, address issues privately.
- Escalate issues to admins (tag @Admins in Discord) for support.

– Assume positive intent

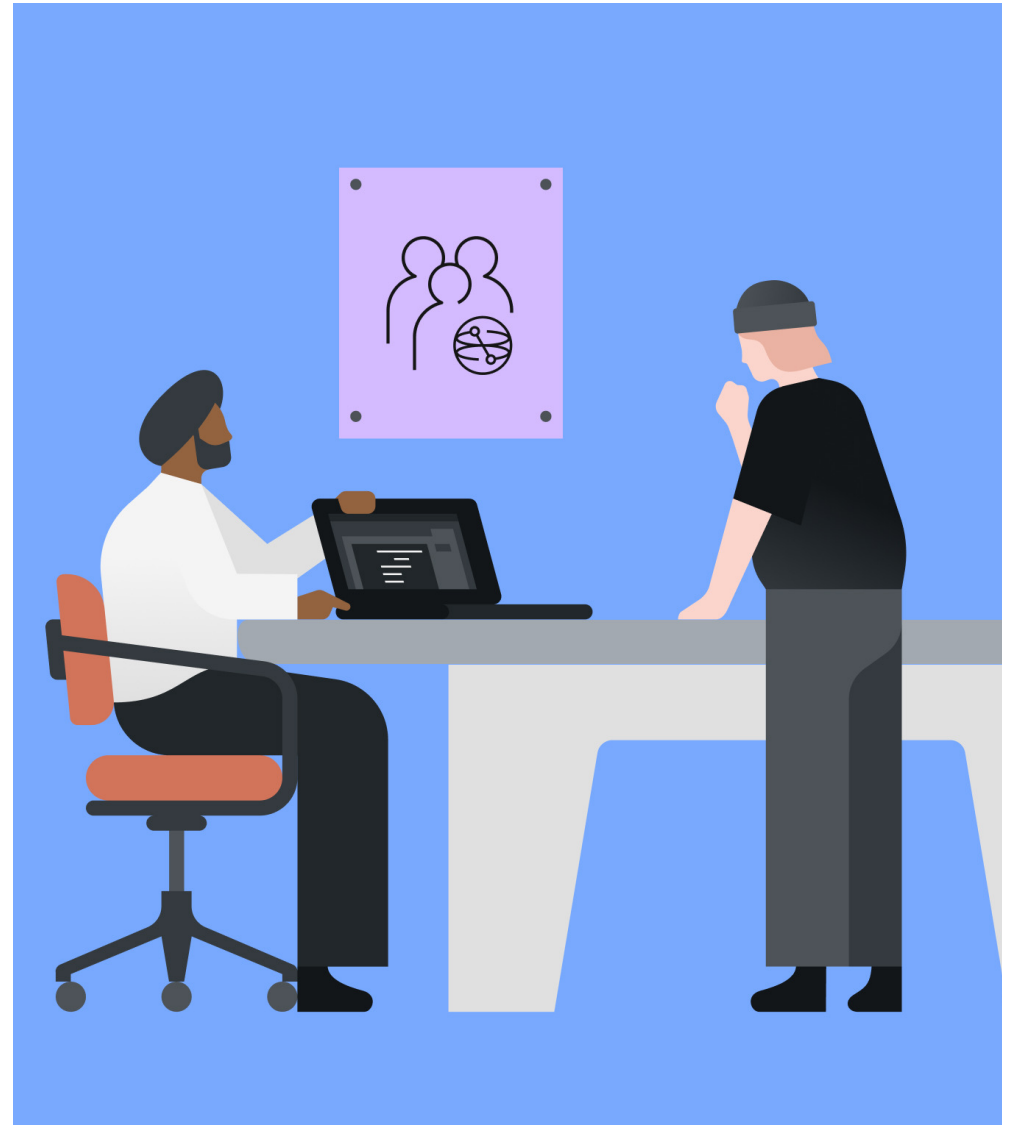
– Use "I" language (e.g. "I feel we're moving too fast")

– Reiterate and realign on the group's shared goals to reunite the team.

– Celebrate the group's progress and achievements, no matter how small, to maintain motivation and morale.

– Uphold the behaviours as outlined in [IBM Quantum's code of conduct](#)

Escalation: *only* when above doesn't work – tag @Admins for mediation assistance with *clearly defined pain points*



How to form a study group...

1. Head to **form-study-groups** and 1) create new post or 2) reply to existing post
 - introduce yourself — share your timezone, current experience level, and when you're hoping to take the exam.
2. Find 2–5 others you want to study with and agree to team up.
Maximum number: 6, Minimum: 3
3. Once your team is ready, the main poster should:
 - a. add “(Closed)” to your post subject
 - b. go to **#join-study-group**, pick an unclaimed team number (SG1, SG2, etc.), and post in the “Claim group numbers below!” thread
4. All group members self-assign the role by clicking the corresponding button in **#join-study-group**. You'll get access to a **private text channel** just for your team to use throughout the 12 weeks.
 - If your team wants a voice channel for meetings, tag admins in your team channel with the request “@Admins we’d like a voice channel please.”

⚠ Please only join a team you actually belong to. Don't self-assign a role for a group you haven't formed with.

You can form groups until **Friday 31 July**.

The screenshot shows the 'form-study-groups' forum channel on the Qiskit Advocate Hub. The main post is titled 'Claim group numbers below!' and was started by Radha Pyari Sandhir. The post includes instructions for forming a study group and a grid of buttons for claiming group numbers (SG1 to SG20). The instructions are as follows:

How to Form Groups

1. Go to the **form-study-groups** forum channel. You can:
 - a) Make a new post and introduce yourself using this template:
Timezone:
When I want to take the exam:
My current Qiskit experience:
OR b) read existing posts and reply to one that matches what you want.
2. When you find 2-5 people you want to study with, agree together to form a group. Send a message in your post on **form-study-groups** "This group is now complete." so others know you are no longer looking for group members.
3. When your group is ready, return to this channel. Look at the **Claim group numbers below!** thread and find the last number used. Claim the next number after it (for example, if SG4 was the last one, claim SG5). Follow the thread's template exactly.
4. Click the button for your group's number to give yourself that role. This gives you access to a private text channel for your team, for the next 12 weeks.

Only click the button for the group you are really part of. Do not click a button for a group you have not joined.

If you accidentally self-assign an incorrect role, simply click the button again to remove the role.

The grid of buttons for claiming group numbers is as follows:

SG1	SG2	SG3	SG4	SG5
SG6	SG7	SG8	SG9	SG10
SG11	SG12	SG13	SG14	SG15
SG16	SG17	SG18	SG19	SG20

Below the grid, there is a message from Radha Pyari Sandhir stating: 'Radha Pyari Sandhir started a thread: Claim group numbers below! See all threads. 16/07/26, 1:16 PM'. Below this, there is a message from Guillaume Saris: 'Guillaume Saris SG14 - @razvicia @Johannes @ilia @Consta... 12h ago'.

What active participation looks like

Active groups meet regularly at a mutually-agreed upon cadence (we suggest weekly, 60–90 min sessions), share the facilitation load, and genuinely engage — coding together, working through practice questions, reviewing concepts. There's no rigid rubric, but admins may check in on your group's channel activity if needed.

If you have already not received one, **Qiskit v2.X Certification exam vouchers are guaranteed for *active participants***, provided all members of the group are in agreement. This is peer-verified, not admin-decided — but inactivity that the whole group agrees to cover up defeats the purpose, and admins may review channel history.

Active participants will be also **eligible for 2 Advocate points** for the activity “Participate in or organize a Quantum Computing club or Qiskit v2.X Certification study group.”

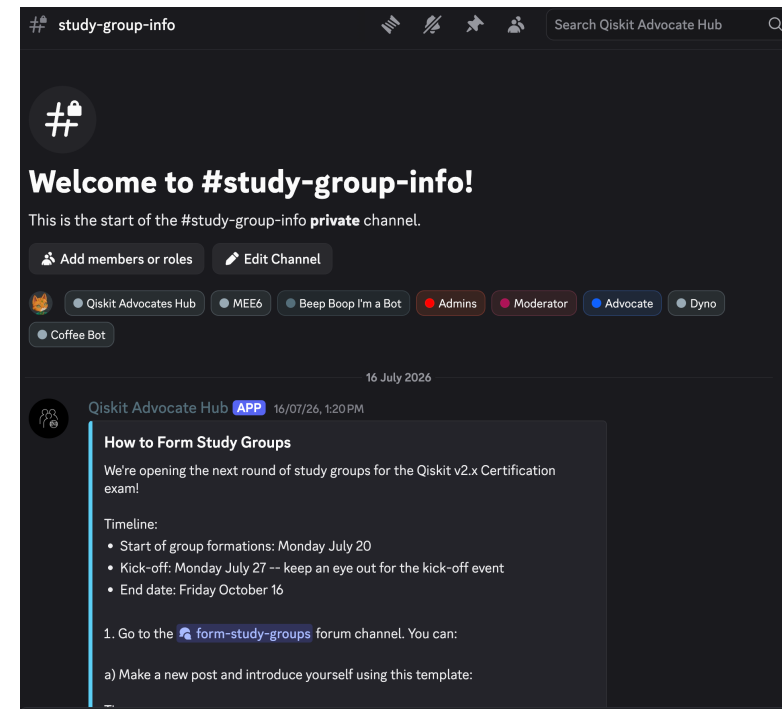
Study group announcements and updates

Announcements channel: #study-group-info

- Announcements
- Updates
- Replay and slide deck of this kick-off session

Timeline:

- Friday July 31 - study group formation ends
- ~Tuesday September 2 – Mid-point check-in
 - Admins will check-in casually to see whether groups are managing well and making progress.
- Friday October 16 – end date



Resources

–Official Qiskit resources

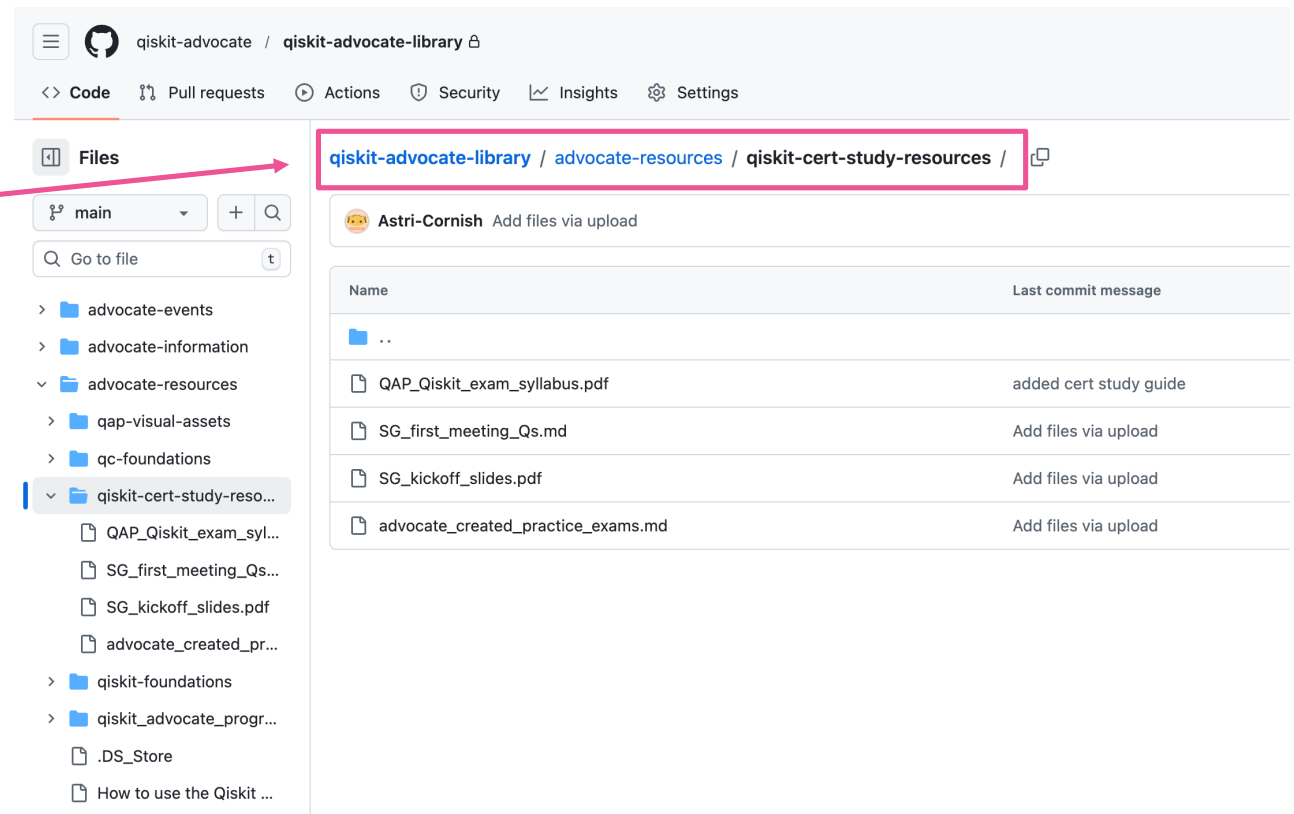
- [IBM Quantum Learning Platform](#)
- [Qiskit Documentation](#)
- [Qiskit tutorials](#)

–[Qiskit advocate library on GitHub](#)

- Exam syllabus with links to resources for each topic
- Suggestions for questions to ask during your first study group meeting & example structure
- Links to Advocate-created practice exams
- This presentation file (with clickable links)

–Consider creating a shared repository for your group

- e.g. a Google Drive folder or a GitHub repository to store and share resources, notes, and code files.
- Other collaboration tools:
 - [Dateful](#): Time conversion across time zones, and more
 - [Doodle](#): Poll for meeting times



Resources

[qiskit-advocate-library](#) / [advocate-resources](#) / [qiskit-cert-study-resources](#) / QAP_Qiskit_exam_syllabus.pdf 

93.4 KB 

Qiskit advocate program

Qiskit v2.X Certification exam guide

Qiskit v2.X Certification Exam Topics	Resources
Perform quantum operations - Define Pauli Operators - Apply quantum operations	Documentation: Operators overview Documentation: Operator class
Visualize quantum circuits, measurements, and states - Visualize quantum circuits - Visualize quantum measurements - Visualize quantum states	Documentation: Visualize circuits Documentation: Visualize results Documentation: Plot quantum states
Create quantum circuits - Construct basic quantum circuits - Construct dynamic circuits - Construct parameterized circuits - Transpile and optimize circuits	Documentation: Construct circuits Documentation: Circuit library Documentation: Classical feedforward and control flow Documentation: Parameterized circuits Video: Introduction to transpilation Documentation: Introduction to Transpilation Documentation: Optimize for hardware Documentation: Set transpiler optimization level
Run quantum circuits - Differentiate execution modes to optimize job queuing - Run quantum circuits with a real hardware in the IBM Runtime provider	Documentation: Execution modes Documentation: Execute on hardware Documentation: Get started with primitives
Use the sampler primitive - Set sampler primitive options such as resilience levels - Bypass runtime error mitigations and implement	Video: An Introduction to Qiskit Runtime Primitives V2 Documentation: Sampler options

Resources

[qiskit-advocate-library](#) / [advocate-resources](#) / [qiskit-cert-study-resources](#) / [SG_first_meeting_Qs.md](#) 

 **Astri-Cornish** Update SG_first_meeting_Qs.md 7b24dcf · 4 months ago  [History](#)

Preview | **Code** | **Blame**

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Questions you may ask during first meeting:

Introductions:

- Introduce yourselves
- Share a bit about your study / work / etc. and why you joined the Qiskit advocate program

Goals and objectives:

- Why does each member want to earn the Qiskit v2.X Certification?
- Apart from obtaining the Qiskit v2.X Certification, what other related goals do the group members have? E.g. create closer connections to other Qiskit advocates, learn from others.

Getting to know your group members' study habits:

- How does each member learn most effectively? If unsure, what have you done in the past regarding learning that has worked for you?
- What keeps each group member motivated?

Study group structure:

- How frequently are you going to meet?
- What days and times are best for the group to meet?
- What content or types of discussions are you going to cover when meeting?
- What are your group members going to do between meetings, if anything?
- What are going to be the roles and responsibilities for each member during and between the meetings? Will these roles and responsibilities be rotated among the group or stay the same over the entire study period?

Resources

Example:

Week	Topic	Session Activities	Homework
1	Introductions, group level-setting, and getting started with Qiskit 2.X	Introductions, discuss prompt questions , assess where group members are at with prior Qiskit and quantum knowledge, decide on meeting cadence and 12-week study plan, install Qiskit, create some basic circuits.	Complete Qiskit installation, review basic Python if needed, review the "perform quantum operations" resources listed in this Qiskit 2.X certification exam syllabus guide .
2	Defining and applying quantum operations	Discuss pauli operators, applying quantum operations, bloch sphere intuition, X, Y, Z, H, S, T, RX/RZ/RZ gates, two-qubit gates like CX, CZ, SWAP, coding session, tackle relevant practice exam questions.	Review the "visualize quantum circuits, measurements, and states" resources listed in this Qiskit 2.X certification exam syllabus guide .
3	Visualizing quantum circuits, measurements, and states	Discuss visualizing quantum circuits, measurements, and states, coding session, tackle relevant practice exam questions.	Review the "create quantum circuits" resources listed in this Qiskit 2.X certification exam syllabus guide .
4	Creating quantum circuits	Discuss creating quantum circuits, coding session, tackle relevant practice exam questions.	Review the "run quantum circuits" resources listed in this Qiskit 2.X certification exam syllabus guide .
5	Running quantum circuits	Discuss running quantum circuits with different execution modes and on real hardware (single jobs), coding session, tackle relevant practice exam questions.	Review the "use the sampler primitive" resources listed in this Qiskit 2.X certification exam syllabus guide .
6	Using the Sampler primitive	Discuss using the Sampler primitive, error mitigation options, resilience levels, coding session, tackle relevant practice exam questions.	Review the "use the estimator primitive" resources listed in this Qiskit 2.X certification exam syllabus guide .

7	Using the Estimator primitive	Discuss using the Estimator primitive, error mitigation options, resilience levels, coding session, tackle relevant practice exam questions.	Review the "retrieve and analyze the results of quantum circuits" resources listed in this Qiskit 2.X certification exam syllabus guide .
8	Retrieving and analyzing the results of quantum circuits	Discuss how to monitor jobs, and retrieve and analyze experiment results, coding session, tackle relevant practice exam questions.	Review the OpenQASM resources listed in this Qiskit 2.X certification exam syllabus guide .
9	Operating with OpenQASM	Discuss OpenQASM semantics, structure types for OpenQASM 3, interoperating between OpenQASM and Qiskit, coding session, tackle relevant practice exam questions.	Review the Qiskit IBM Runtime REST API resources listed in this Qiskit 2.X certification exam syllabus guide .
10	Interacting with the Qiskit IBM Runtime REST API	Discuss using the Qiskit IBM Runtime REST API, tackle relevant practice exam questions.	Complete a full-length mock exam under timed conditions, note down areas of weakness or difficult questions.
11	Mock exam review	Review solutions of the mock exam together, discuss difficulties, share approaches to difficult questions.	Review exam practice questions and topic resources in areas of weakness.
12	Final review & certification readiness	Lightning review of all exam topics, discussions on most difficult concepts or example questions, exam strategy, confidence check.	Submit study group finalization form and sit the certification exam (or schedule it within the following weeks).

Each group meeting may look like:

1. 10 min: Review of topic content/resources

- From the homework, discuss common mistakes or hard-to-understand concepts.

2. 20 min: Concept lesson

- Review key theory of this week's focus topic, which may include writing explanations, key definitions, and key concepts on a shared group (perhaps virtual) whiteboard.

3. 30 min: Hands-on coding

- Attempt coding examples and exercises from the topic content, play with code, create your own code examples on the topic.

4. 20 min: Certification practice questions

- Tackle practice questions that relate to the week's topic from this bank of [Advocate-created practice exams](#).

5. 10 min: Wrap-up

- Key takeaways
- Homework
- Resources
- Confirm next meeting time and date

Resources

qiskit-advocate-library / advocate-resources / qiskit-cert-study-resources / advocate_created_practice_exams.md

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Qiskit advocate-created practice exams for the [Qiskit v2.X Developer Certification](#) exam:

Link to practice exam	Creator's Discord username
ert-practice-exam/blob/main/practice-exam-1.md	clausia
nputingMeetup/qiskit_developer_certification/blob/main/Sample_Exam_Qiskit2.0_Developer_Certification_2025.pdf	nhawkins_seattle
kitAdvocate2025Assignments/blob/main/QiskitPracticeExam.pdf	rishabron
it-certification-prep	dmurata
y/tree/main/Qiskit%20v2.x%20Certification	the_youri_
_qiskit_2_x	kurianuthuppu
certification-practice-exam	rishihg
-v2.X-developer-certification-practice-test	luke_j_miller
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amGen	feras6399
-Certification-Practice-Questions/blob/main/Qiskit-Practice-Exam.ipynb	lukakaucic_37602
m-quiz	maja
	ns_bs

Resources

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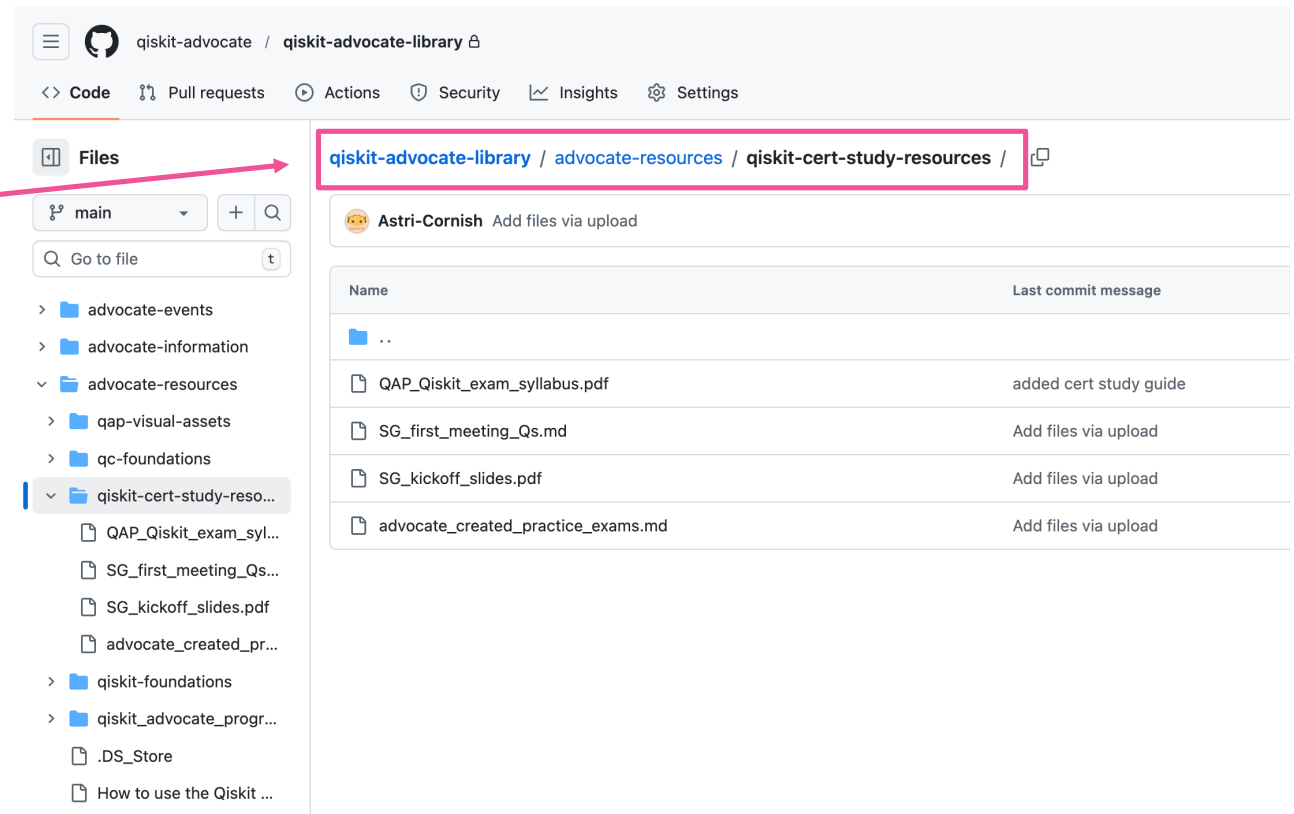
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Questions?

